INCH-POUND

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MILITARY SPECIFICATION

CABLE ASSEMBLY, POWER, ELECTRICAL: ABCA-M168

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electrical power cable assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

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STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials

MILITARY

MIL-STD-454-General Requirements for Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-15-5550 - Cable Assembly, Power, Electrical: ABCA-M168

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-15-5550 - Cable Assembly, Power, Electrical: ABCA-M168

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CTF, Rock Island, IL 61299-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-15-5550 and on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings and in this specification.

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3.1.2 Components. All components of the cable assembly shall conform to the specifications and drawings listed on Drawing 5-15-5550 and subsidiary drawings.

3.2 Manufacture and assembly. The components shall be manufactured and assembled as specified on the subsidiary drawings listed on Drawing 5-15-5550.

3.3 Resistance of cable assembly. The cable assembly shall have the resistances specified in table I when tested in accordance with 4.4.6.1.

Table I. Resistance between test points

From	To	Resistance (ohms)
P1-A	P1-E	1.0, max.
P1-A	P2-5	1.0, max.
P1-B	P2-1	1.0, max.
P2-2	P2-4	1.0, max.
P1-Shell	P1-A	1.0×10^6 min.
P1-Shell	P1-B	1.0×10^6 min.
P2-1	P2-5	1.0×10^6 min.
P2-Shell	P2-1	1.0×10^6 min.
P2-Shell	P2-2	1.0×10^6 min
P2-Shell	P2-5	1.0×10^6 min.

3.4 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection (see 6.3) in accordance with 4.3.

3.5 Workmanship. The workmanship of the cable assembly shall be in accordance with MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all

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products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.5) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Sample. The first article sample shall consist of three cable assemblies manufactured and packaged (when applicable) using the same methods, materials, equipment, and processes as will be used during regular production and three empty unit pack containers. Empty unit pack containers shall not be part of the first article sample when packaging is for interplant shipment. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

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4.3.3 Rejection criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and at essentially the same time.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table II. Samples shall be selected at random.

TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

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4.4.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(a)	Cable Assembly, Power, Electrical: ABCA-M168		5-15-5550
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Resistance of cable assembly	100% inspection	3.3	CE
102	Workmanship	Table II, level III	3.5	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Packaging		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Unit pack container leakage	Table II, level III	5.1	4.4.7.1
102	Unit pack container seam	Acc 0, Rej 1*	5.1	4.4.7.2**
103	Item completely cleaned and dry prior to unit packing	Table II, level III	5.1	VI
104	Intimate wrap evident and correct	Table II, level III	5.1	VI and CE
105	Unit pack container evident and correct	Table II, level III	5.1	VI and CE
106	Supplemental container evident and correct	Table II, level III	5.1	VI and CE
107	Unit pack container markings correct evident and legible	Table II, level III	5.1	VI and CE

NOTES:
This paragraph is not applicable to packaging for interplant shipment.
* Sample shall consist of five closed empty unit pack containers
** Destructive test
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests. The test on the cable assembly shall be conducted as follows:

4.4.6.1 Resistance of cable assembly. Connect an ohmmeter with an accuracy of no less than ± 2°of arc on the R x 1 scale, 1.5° of arc on the R x 100 scale and R x 10K scale, successively to each pair of test points listed in table I. Measure for the specified resistance at each pair of test points.

4.4.7 Packaging tests.

4.4.7.1 Unit pack container leakage. The unit pack container shall be tested in accordance with method 5009.2 of FED. TEST METHOD STD. NO. 101 using the hot water technique.

4.4.7.2 Unit pack container seam. The closure seam of the empty unit pack container shall be prepared under the same conditions used for unit pack assemblies. The closure seam of the empty unit pack container shall be tested in accordance with method 2024 of FED. TEST METHOD STD. NO. 101, except that the static load shall be 45 (+5 or -0) ounces and a 5 percent reduction in the static load shall be permitted when the room temperature exceeds 90°F in the test area.

5. PACKAGING

5.1 Interplant shipment (see 6.4). Military packaging as specified in 5.2 is not applicable to interplant shipment. The cable assembly shall be packaged to provide adequate protection from physical damage, and marked to identify the cable assembly and provide any other required data during shipment from the source to the first receiving activity.

5.2 Military packaging. For all shipments, other than interplant shipments (see 6.4) packaging (preservation, unit packing and marking) shall be as specified in SPI P5-15-5550.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cable assembly is intended to connect the detector unit of the M8A1 Chemical Agent Automatic Alarm to a second battery. The connector at the battery end of the cable assembly is especially designed to be highly resistant to the electrolyte used in the secondary battery.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification

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(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Whether military packaging or interplant shipment packaging is required

(e) Level of packing required

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.4 Interplant shipment. Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from vendor to a subcontractor or a prime contractor, or between contractors and subcontractors, or from a vendor to a military arsenal, plant or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.5 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.6 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through the inter-

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national standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.7 Subject term (key word) listing.

Cable electrical
Connector, electrical cable
Electrical power cable

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6665-A610

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*		
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6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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